

Project Documentation: Nutrition Assistant

1. Introduction

Project Title: Nutrition Assistant

Team Members:

Name :- BITRA SAI HEMA LATHA

Role: Full Stack Developer (Frontend, Backend, Database, Testing & Deployment)

2. Project Overview

Purpose: Nutrition Assistant is a full-stack wellness management system that helps users and dietitians create, manage, and track personalized nutrition plans through an interactive interface.

Features

- User Registration & Login
- JWT Authentication
- Dashboard with Daily Calorie Tracker
- Macronutrient Tracking
- Add/Edit/Delete Meals
- Water Intake Tracking
- Exercise Tracking
- Profile Management
- Daily Goal Management
- Nutrition History
- Responsive User Interface

3. Architecture

Frontend (React.js) :-

The frontend is built using React.js with reusable components. React Router manages navigation between pages while Axios communicates with backend REST APIs.

Technologies

- React.js
- HTML5
- CSS3
- JavaScript
- Axios
- React Router

Backend (Node.js & Express.js) :-

The backend handles authentication, API routing, business logic, nutrition calculations, and profile management.

Technologies

- Node.js
- Express.js
- JWT Authentication
- bcryptjs
- Mongoose

Database (MongoDB) :-

MongoDB stores

- User Details
- User Profiles
- Meal Logs
- Nutrition Logs
- Daily Goals

- Water Intake
- Exercise Records

Collections

- Users
- NutritionLogs

4. Setup Instructions

Prerequisites:

- Node.js
- MongoDB
- VS Code
- Git
- Npm

.

Installation:

Clone Repository

git clone:- <https://github.com/hema6526/Nutrition-Assistant.git>

Install Frontend:-

```
cd client  
npm install
```

Install Backend:-

```
cd server  
npm install
```

Environment Variables (.env)

PORT=5000

MONGO_URI= mongodb://localhost:27017/nutrition-assistant

JWT_SECRET= your_super_secret_jwt_key_change_this

5. Folder Structure:-

```
NutritionAssistant/  
  
├─ client/  
│  
├─ public/  
│  
├─ src/  
│   ├─ components/  
│   ├─ pages/  
│   ├─ services/  
│   ├─ App.js  
│   └─ index.js  
│  
├─ server/  
│  
├─ config/  
├─ controllers/  
├─ middleware/  
├─ models/  
├─ routes/  
├─ utils/  
├─ server.js  
└─ package.json
```

6. Running the Application

- **Frontend:-**
cd client
npm start
- **Backend:-**
cd server
npm start
- **Application URLs:-**

Frontend:-
<http://localhost:3000>

- **Backend :-**

http://localhost:5000

7. API Documentation

Method	Endpoint	Purpose
POST	/api/auth/login	User/Dietitian authentication
GET	/api/clients/profile	Fetch client health stats
POST	/api/meals/create	Dietitian creates a meal plan
PUT	/api/progress/update	Update daily nutrient adherence

8. Authentication

Roles (User, Dietitian, Administrator) are managed via JWT. Passwords are encrypted with bcrypt. The system restricts administrative functions (approvals, data analytics) to specific roles :-

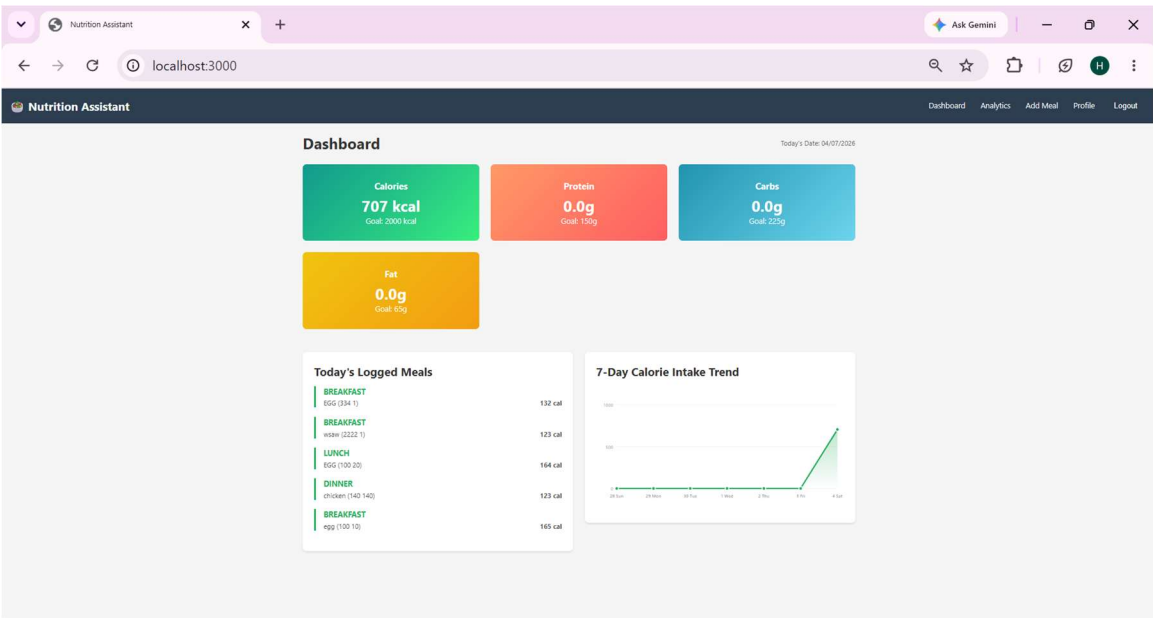
The application uses JWT Authentication.

Workflow

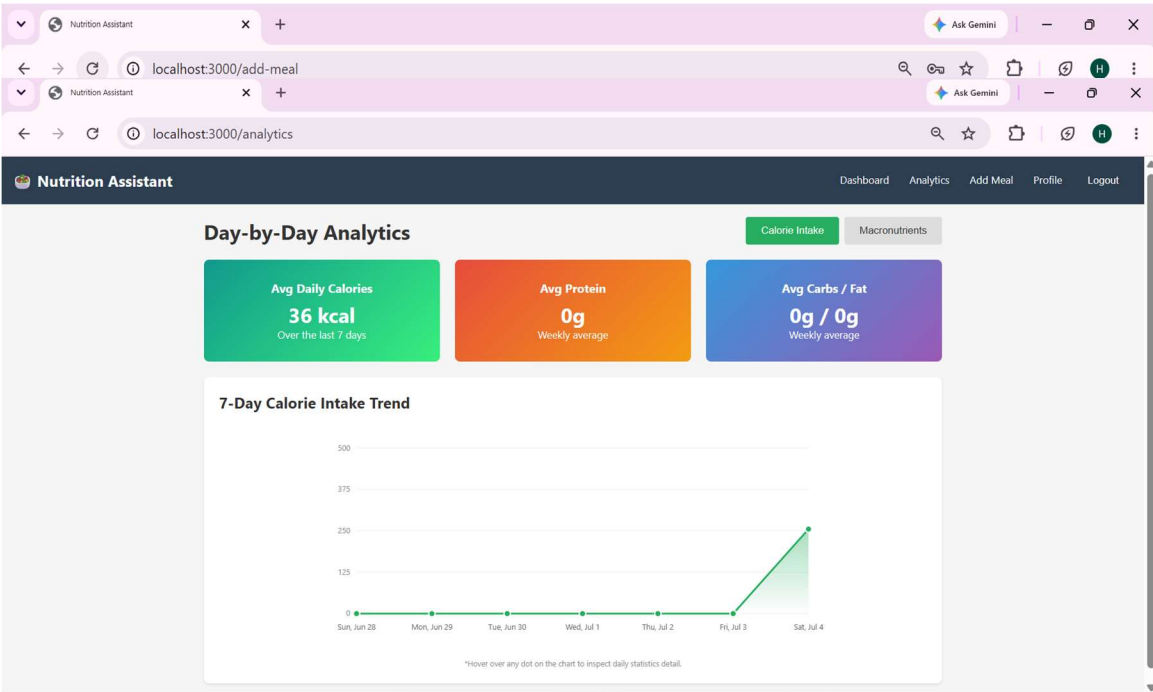
1. User Registers
2. Password encrypted using bcryptjs
3. JWT Token generated
4. Token stored on client
5. Every API request verifies JWT Token
6. Authorized users access protected pages

9. User Interface

Placeholder for Dashboard



MEAL PLANNING FORMS



Analytics :-

10. Testing

API testing via Postman. Integration testing for CRUD workflows. UI testing ensures cross-browser compatibility :-

Testing Performed

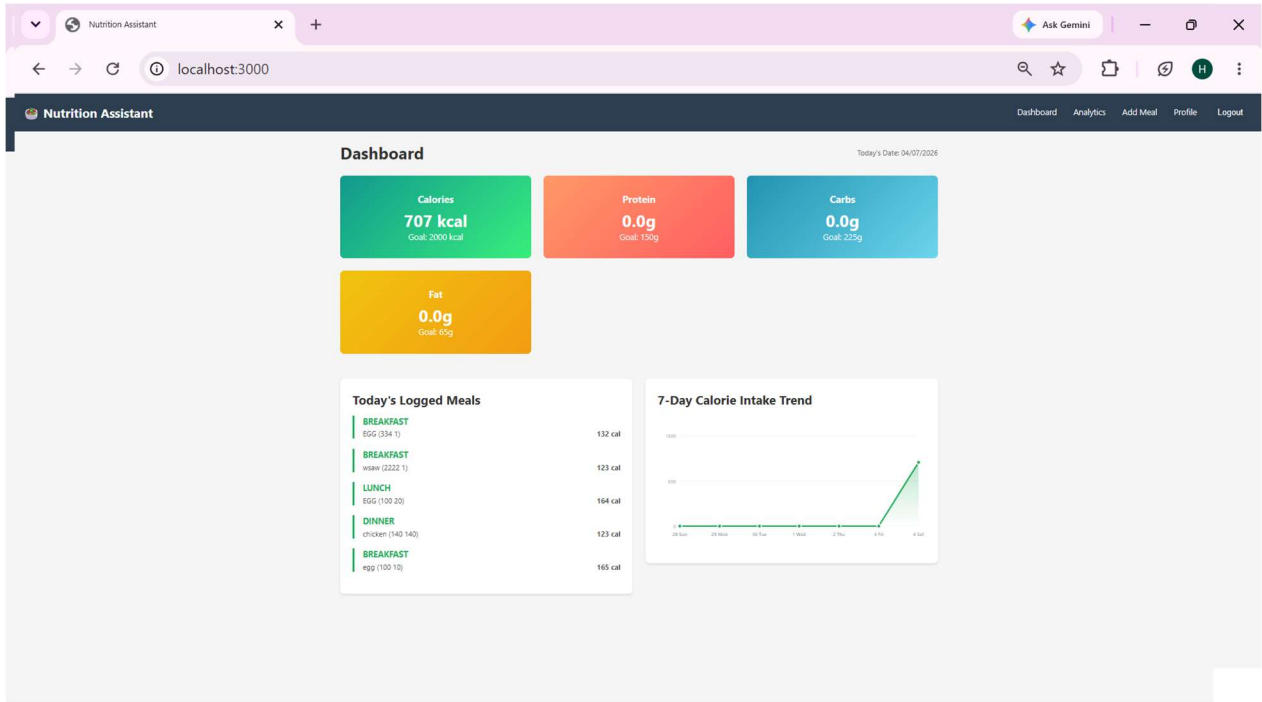
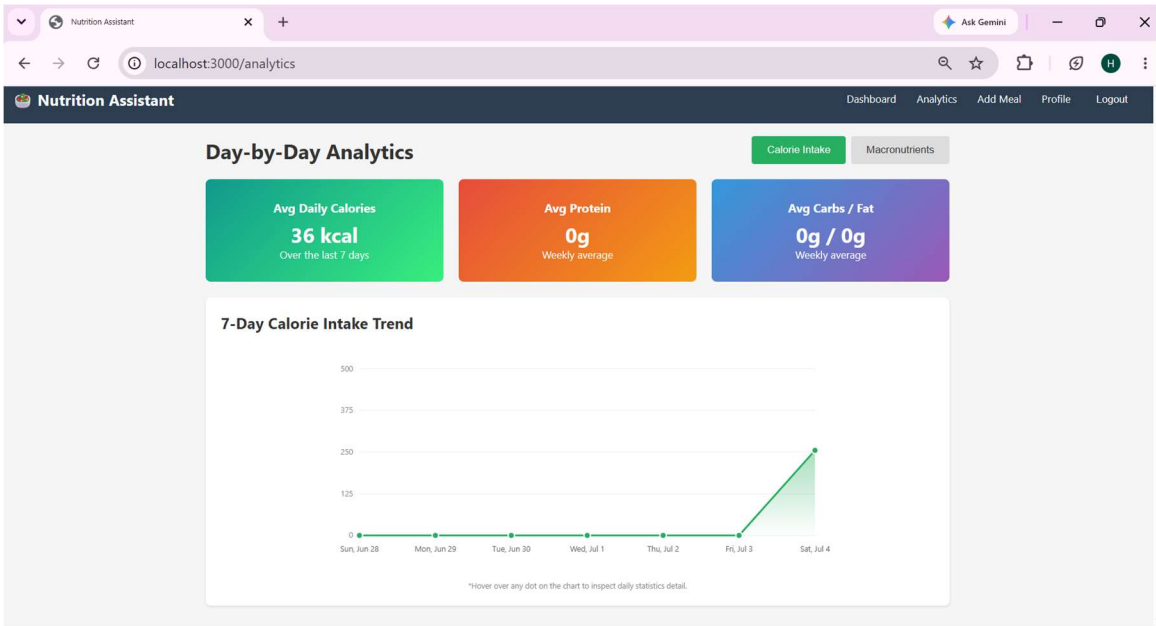
- User Registration
- Login Authentication
- JWT Verification
- Add Meal
- Dashboard Update
- Profile Update
- Logout

Testing Type

- Functional Testing
- Integration Testing
- Manual Testing

11. Screenshots or Demo

DEMO LINK & SCREENSHOTS :- (<http://localhost:3000>)



12. Known Issues

- Potential latency in historical chart rendering (Optimization ongoing)
- Internet connection required for cloud deployment.
- No barcode food scanner available.
- No AI meal recommendations.
- No mobile application version.
- Limited report visualization.

13. Future Enhancements

- By completing this project, you will:
- Build a real-world full-stack web application using the MERN stack.
- Implement role-based authentication and dynamic nutrition tracking.
- Develop a responsive, data-driven dashboard using React.
- Gain experience in backend design, API integration, and secure authentication.
- Understand modular development and performance optimization for large-scale wellness platforms.
- AI-driven meal recognition.
- Barcode scanning for instant nutritional lookups.
- Integration with Fitbit/Apple Watch.